Sunrise Diagnostics

Plot 12, Baner Road, Pune 411045
Patient Name: Mr. Arjun Menon
Age / Sex: 66 Years / Male
Lab ID: SD-25-092011
Ref. By: Dr. Anil Deshpande
Collected On: 20/09/2025 07:50 AM
Reported On: 20/09/2025 04:10 PM

HAEMATOLOGY

Test	Result	Flag	Unit	Reference Range
Haemoglobin	13.8		g/dL	13.0 - 17.0
RBC Count	4.7		mill/cumm	4.5 - 5.5
PCV	42.1		%	40.0 - 50.0
MCV	89.6		fL	83 - 101
MCH	29.4		pg	27.0 - 32.0
MCHC	32.8		g/dL	31.5 - 34.5
RDW-CV	13.1		%	11.6 - 14.0
Platelet Count	2,18,000		/cumm	1,50,000 - 4,10,000
Total WBC Count	8,300		/cumm	4,000 - 10,000
Neutrophils	61		%	40 - 80
Lymphocytes	29		%	20 - 40
Eosinophils	5		%	1 - 6
Monocytes	4		%	2 - 10
Basophils	1		%	0 - 2
ESR	12		mm/hr	0 - 15

GLUCOSE AND LIPIDS

Test	Result	Flag	Unit	Reference Range
Fasting Plasma Glucose	108	H	mg/dL	70 - 100
HbA1c	6.0		%	Normal: < 5.7; Prediabetes: 5.7 - 6.4; Diabetes: >= 6.5
Total Cholesterol	168		mg/dL	Desirable: < 200; Borderline: 200 - 239; High: >= 240
Triglycerides	142		mg/dL	< 150
HDL Cholesterol	38	L	mg/dL	> 40
LDL Cholesterol	94		mg/dL	Optimal: < 100; Near optimal: 100 - 129; Borderline high: 130 - 159; High: >= 160
VLDL Cholesterol	28.4		mg/dL	< 30

KIDNEY FUNCTION TEST

Test	Result	Flag	Unit	Reference Range
Blood Urea	31		mg/dL	15 - 40
Blood Urea Nitrogen (BUN)	14.5		mg/dL	7 - 20
Serum Creatinine	1.0		mg/dL	0.7 - 1.3
Uric Acid	6.1		mg/dL	3.5 - 7.2
eGFR	84		mL/min/1.73m²	> 60

Synthetic test document: fictional patient, doctors and laboratory.

LIVER FUNCTION TEST

Test	Result	Flag	Unit	Reference Range
Total Bilirubin	0.9		mg/dL	0.3 - 1.2
Direct Bilirubin	0.2		mg/dL	< 0.3
Indirect Bilirubin	0.7		mg/dL	0.2 - 0.8
SGOT (AST)	28		U/L	< 40
SGPT (ALT)	34		U/L	< 41
Alkaline Phosphatase	88		U/L	40 - 129
GGT	31		U/L	< 60
Total Protein	7.3		g/dL	6.4 - 8.3
Albumin	4.4		g/dL	3.5 - 5.2
Globulin	2.9		g/dL	2.0 - 3.5
A/G Ratio	1.5			1.1 - 2.5

ELECTROLYTES AND MINERALS

Test	Result	Flag	Unit	Reference Range
Sodium	140		mmol/L	135 - 145
Potassium	4.2		mmol/L	3.5 - 5.1
Chloride	103		mmol/L	98 - 107
Magnesium	2.1		mg/dL	1.6 - 2.6
Calcium	9.4		mg/dL	8.6 - 10.2
Phosphorus	3.6		mg/dL	2.5 - 4.5

SPECIAL TESTS

Test	Result	Flag	Unit	Reference Range
T3 (Total)	1.2		ng/mL	0.8 - 2.0
T4 (Total)	7.8		µg/dL	5.1 - 14.1
TSH	2.1		µIU/mL	0.27 - 4.20
Serum Iron	96		µg/dL	65 - 175
TIBC	318		µg/dL	250 - 450
Transferrin Saturation	30.2		%	20 - 50
Serum Ferritin	142		ng/mL	30 - 400
Vitamin D (25-OH)	34.2		ng/mL	Deficiency: < 20; Insufficiency: 20 - 30; Sufficiency: 30 - 100
Vitamin B12	412		pg/mL	211 - 911
Total PSA	1.4		ng/mL	< 4.0

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URINE ROUTINE

Test	Result	Flag	Unit	Reference Range
Colour	Pale yellow			
Appearance	Clear			
Specific Gravity	1.015			1.005 - 1.030
Protein	Nil			Nil
Glucose	Nil			Nil
Pus Cells	1-2		/hpf	0 - 5

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TEST MENU AND METHODS

Code	Test	Sample	Vol.	Stability	TAT	Fasting	Runs	Department	Accredited	Rs	Method
HEM-001	Haemoglobin	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	180	SLS-haemoglobin photometry
HEM-002	RBC Count	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	210	electrical impedance with hydrodynamic focusing
HEM-003	PCV	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	240	calculated from the RBC count and the MCV
HEM-004	MCV	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	270	electrical impedance
HEM-005	MCH	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	300	calculated from haemoglobin and the RBC count
HEM-006	MCHC	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	330	calculated from haemoglobin and the PCV
HEM-007	RDW-CV	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	150	calculated from the red cell volume histogram
HEM-008	Platelet Count	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	180	electrical impedance
HEM-009	Total WBC Count	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	210	flow cytometry with fluorescent nucleic acid staining
HEM-010	Neutrophils	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	240	flow cytometry (five-part differential); a smear is reviewed when the analyser flags
HEM-011	Lymphocytes	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	270	flow cytometry (five-part differential)
HEM-012	Eosinophils	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	300	flow cytometry (five-part differential)
HEM-013	Monocytes	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	330	flow cytometry (five-part differential)
HEM-014	Basophils	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	150	flow cytometry (five-part differential)
HEM-015	ESR	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	180	modified Westergren method
GLU-001	Fasting Plasma Glucose	Fluoride (grey)	2 mL	24 h, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	150	hexokinase
GLU-002	HbA1c	EDTA (lavender)	2 mL	24 h, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	180	ion-exchange HPLC
GLU-003	Total Cholesterol	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	210	cholesterol oxidase-peroxidase (CHOD-PAP)
GLU-004	Triglycerides	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	240	glycerol phosphate oxidase-peroxidase (GPO-PAP)
GLU-005	HDL Cholesterol	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	270	direct homogeneous enzymatic method
GLU-006	LDL Cholesterol	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	300	calculated by the Friedewald equation when triglycerides are below 400 mg/dL
GLU-007	VLDL Cholesterol	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	120	calculated as triglycerides divided by five
KFT-001	Blood Urea	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	No	Daily	Clinical Biochemistry	Yes	210	urease-glutamate dehydrogenase
KFT-002	Blood Urea Nitrogen (BUN)	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	No	Daily	Clinical Biochemistry	Yes	240	calculated as urea divided by 2.14
KFT-003	Serum Creatinine	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	No	Daily	Clinical Biochemistry	Yes	270	modified kinetic Jaffe method
KFT-004	Uric Acid	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	No	Daily	Clinical Biochemistry	Yes	300	uricase-peroxidase
KFT-005	eGFR	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	No	Daily	Clinical Biochemistry	Yes	330	calculated by the CKD-EPI 2021 creatinine equation
LFT-001	Total Bilirubin	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	230	diazo method with an accelerator
LFT-002	Direct Bilirubin	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	260	diazo method without an accelerator
LFT-003	Indirect Bilirubin	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	290	calculated as total minus direct bilirubin
LFT-004	SGOT (AST)	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	320	IFCC method without pyridoxal phosphate
LFT-005	SGPT (ALT)	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	350	IFCC method without pyridoxal phosphate
LFT-006	Alkaline Phosphatase	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	380	IFCC method
LFT-007	GGT	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	200	IFCC method (Szasz)
LFT-008	Total Protein	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	230	biuret method
LFT-009	Albumin	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	260	bromocresol green
LFT-010	Globulin	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	290	calculated as total protein minus albumin
LFT-011	A/G Ratio	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	320	calculated as albumin divided by globulin
ELM-001	Sodium	Serum (gold)	3 mL	7 d, 2-8 °C	4 h	No	Daily	Clinical Biochemistry	Yes	250	indirect ion-selective electrode
ELM-002	Potassium	Serum (gold)	3 mL	7 d, 2-8 °C	4 h	No	Daily	Clinical Biochemistry	Yes	280	indirect ion-selective electrode
ELM-003	Chloride	Serum (gold)	3 mL	7 d, 2-8 °C	4 h	No	Daily	Clinical Biochemistry	Yes	310	indirect ion-selective electrode
ELM-004	Magnesium	Serum (gold)	3 mL	7 d, 2-8 °C	4 h	No	Daily	Clinical Biochemistry	Yes	340	xylidyl blue
ELM-005	Calcium	Serum (gold)	3 mL	7 d, 2-8 °C	4 h	No	Daily	Clinical Biochemistry	Yes	370	arsenazo III
ELM-006	Phosphorus	Serum (gold)	3 mL	7 d, 2-8 °C	4 h	No	Daily	Clinical Biochemistry	Yes	400	phosphomolybdate
SPL-001	T3 (Total)	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	480	chemiluminescent immunoassay (CLIA)
SPL-002	T4 (Total)	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	510	chemiluminescent immunoassay (CLIA)
SPL-003	TSH	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	540	third-generation chemiluminescent immunoassay (CLIA)
SPL-004	Serum Iron	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	570	ferrozine method
SPL-005	TIBC	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	600	direct method with magnesium carbonate adsorption
SPL-006	Transferrin Saturation	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	630	calculated as serum iron divided by TIBC
SPL-007	Serum Ferritin	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	450	particle-enhanced immunoturbidimetric assay
SPL-008	Vitamin D (25-OH)	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	480	chemiluminescent immunoassay (CLIA)
SPL-009	Vitamin B12	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	510	competitive chemiluminescent immunoassay (CLIA)
SPL-010	Total PSA	Serum (gold)	3 mL	7 d, 2-8 °C	Next day	No	Mon-Sat	Immunoassay	Yes	540	chemiluminescent immunoassay (CLIA)
URN-001	Colour	Urine container	As supplied	See note	2 h	No	Daily	Clinical Pathology	Yes	130	visual examination
URN-002	Appearance	Urine container	As supplied	See note	2 h	No	Daily	Clinical Pathology	Yes	160	visual examination
URN-003	Specific Gravity	Urine container	As supplied	See note	2 h	No	Daily	Clinical Pathology	Yes	190	reagent strip (ion exchange)
URN-004	Protein	Urine container	As supplied	See note	2 h	No	Daily	Clinical Pathology	Yes	220	reagent strip
URN-005	Glucose	Urine container	As supplied	See note	2 h	No	Daily	Clinical Pathology	Yes	250	reagent strip (glucose oxidase)
URN-006	Pus Cells	Urine container	As supplied	See note	2 h	No	Daily	Clinical Pathology	Yes	280	microscopy of the centrifuged deposit
COA-001	Prothrombin Time (PT/INR)	Citrate (blue)	2.7 mL	4 h, room temp.	4 h	No	Daily	Haematology	Yes	380	clotting time, photo-optical
COA-002	aPTT	Citrate (blue)	2.7 mL	4 h, room temp.	4 h	No	Daily	Haematology	Yes	410	clotting time, photo-optical
COA-003	Fibrinogen	Citrate (blue)	2.7 mL	4 h, room temp.	4 h	No	Daily	Haematology	Yes	440	Clauss method
COA-004	D-Dimer	Citrate (blue)	2.7 mL	4 h, room temp.	4 h	No	Daily	Haematology	Yes	470	latex immunoturbidimetry
HEM-016	Blood Group and Rh Type	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	210	column agglutination
HEM-017	Peripheral Smear Examination	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	240	Leishman stain, microscopy
HEM-018	Reticulocyte Count	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	270	flow cytometry
HEM-019	Malaria Antigen	EDTA (lavender)	2 mL	24 h, 2-8 °C	4 h	No	Daily	Haematology	Yes	300	immunochromatography
GLU-008	Post-prandial Plasma Glucose	Fluoride (grey)	2 mL	24 h, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	150	hexokinase
GLU-009	Random Plasma Glucose	Fluoride (grey)	2 mL	24 h, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	180	hexokinase
GLU-010	Oral Glucose Tolerance Test	Fluoride (grey)	2 mL	24 h, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	210	hexokinase, 0 and 120 min
GLU-011	Apolipoprotein B	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	240	immunoturbidimetry
GLU-012	Lipoprotein (a)	Serum (gold)	3 mL	7 d, 2-8 °C	Same day	10-12 h	Daily	Clinical Biochemistry	Yes	270	immunoturbidimetry
KFT-006	Cystatin C	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	No	Daily	Clinical Biochemistry	Yes	360	particle-enhanced immunoturbidimetry
KFT-007	Urine Albumin/Creatinine Ratio	Spot urine	As supplied	See note	6 h	No	Daily	Clinical Biochemistry	Yes	180	immunoturbidimetry and Jaffe
KFT-008	24-hour Urine Protein	24-hour urine	As supplied	See note	6 h	No	Daily	Clinical Biochemistry	Yes	210	pyrogallol red
LFT-012	Serum Amylase	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	350	CNPG3 substrate, kinetic
LFT-013	Serum Lipase	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	380	colorimetric, kinetic
LFT-014	LDH	Serum (gold)	3 mL	7 d, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	200	lactate to pyruvate, kinetic UV
LFT-015	Serum Ammonia	EDTA on ice	2 mL	24 h, 2-8 °C	6 h	Preferred	Daily	Clinical Biochemistry	Yes	230	glutamate dehydrogenase, UV

All results relate only to the sample as received. Reference intervals are those of this laboratory for adults and may differ from other laboratories.
A result outside its reference interval is not by itself a disease; about one healthy person in twenty falls outside any single interval.
Results are to be interpreted by a registered medical practitioner, together with the history, the examination and other investigations.
Values flagged H or L lie above or below the printed reference interval. Critical values are telephoned to the referring doctor.

Dr. Neha Joshi, MD (Pathology)

*** End of Report ***

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